SMART WORKWEAR SYSTEM

User Manual

Ergonomic risk assessment & work technique training

Manual version 3.0
Mobile app Android v2.9.x
Applies to Wergonic Mobile App & Web Dashboard

Contents

1. The Wergonic System

Wergonic is a smart-workwear system for fast, accurate measurement of upper-body posture and motion during real work tasks — with results compared directly against recommended reference values.

Smart Workwear

- Motion sensors fitted into garments for on/off use
- Tracks trunk & head, arms, and wrist
- Estimates reachability (hand's reach)

Mobile App (Android)

- Streams and records sensor data over Bluetooth
- Real-time posture & movement metrics
- Annotate activities and breaks
- Syncs sessions to the web dashboard

Web Dashboard

- Programme activity & work-cycle tags
- Generate risk-assessment reports
- Built-in frameworks: RAMP, RULA, REBA, MEC (Volvo)
- AI-assisted interpretation of results
- User & organisation admin tools
- Runs in any browser — **cloud.wergonic.com**

> **Your account.** You receive credentials to sign in to both the app and the dashboard with an email of your choice. Within an organisation, colleagues' accounts are linked for shared work.

2. What's in the Kit

The Wergonic Smart Workwear is delivered as a standardised kit.

- **Six IMUs** in colour-coded moulds
- **Android phone** with the Wergonic App installed and pre-paired
- **Wergonic shirts** in six sizes (XS–XXL)
- **Wergonic glove** and **headband**
- **Six CR2025** coin-cell batteries and one **phone charger**
- **Heart-rate chest band** with an extra mould-free IMU (optional)

Phone compatibility

For best performance use an Android phone on **Android 13 or newer**. Bluetooth hardware quality affects data streaming. Thoroughly tested models:

- Samsung Galaxy S21 FE · Samsung Galaxy S21 Ultra
- Google Pixel 6A · Google Pixel 7A

Compliance: CE, FCC, IC · Bluetooth 4.0/5.0 · RoHS, REACH.

3. Sensors & What It Measures

The system uses six Movesense inertial measurement units (IMUs) held in colour-coded moulds for correct placement.

Sensor & specs

- 9-axis motion (accelerometer, gyroscope, magnetometer) + temperature
- Accelerometer ±2/4/8/16 g · Gyroscope ±125–2000 °/s
- Sampling 12.5–833 Hz · Bluetooth 4.0/5.0
- 36.6 mm ⌀ × 10.6 mm · 9.4 g · CR2025 battery
- Waterproof & shockproof

Placement & colour code

- Right arm — Blue
- Trunk — Green
- Head — Black
- Left arm — Red
- Hand — Silver
- Forearm — Beige

One IMU on each arm (deltoid), one on the trunk (cervical/thoracic junction), one on the head (forehead), one on the dominant forearm and one on the dominant hand.

Measured quantities

Trunk & head inclination	Angle (°) of trunk / head relative to a vertical reference set during calibration — flexion, extension or a combination.
Arm elevation	Upper-arm posture (°) and speed of movement (°/s) — all shoulder movements relative to vertical.
Wrist angular velocity	Rate of rotation (°/s) between hand and forearm, from the hand and forearm IMUs.

4. System Setup

You can use the provided phone (mostly pre-configured) or your own Android phone. Either way, finish by creating your tags on the dashboard.

4.1 · Using the provided phone

The kit phone already has the correct Android version, the latest Wergonic App, and pre-paired IMUs. Two steps remain:

1. **Sign in to Google Play** with a Gmail account so the app stays updated with new releases.
2. **Create your tags** on the dashboard (see 4.4).

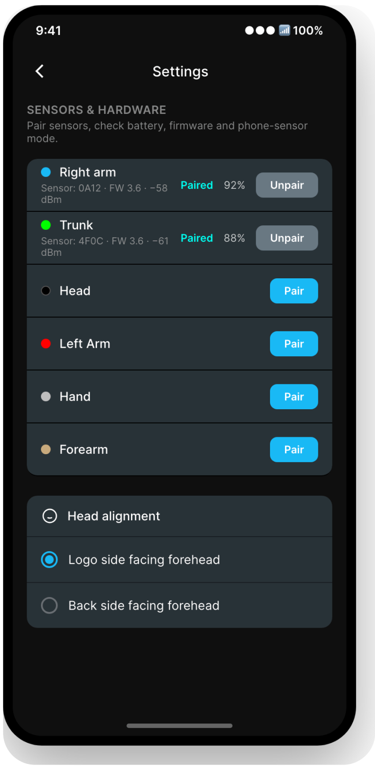
Resetting the phone to factory settings is optional. If you do, follow the "own phone" steps below afterwards.

4.2 · Using your own phone

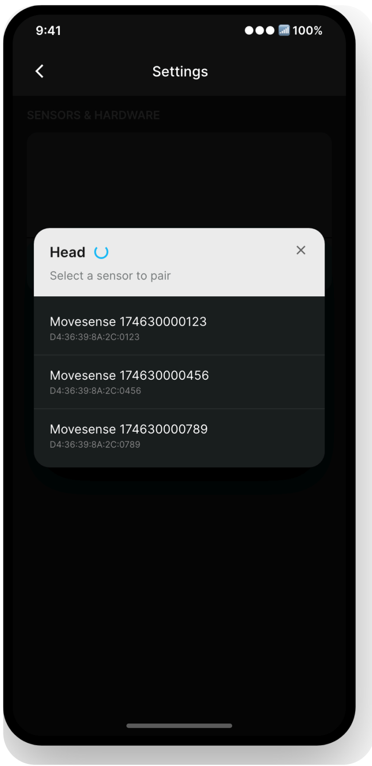
1. **Install the app** — open Google Play, sign in, search **Wergonic**, install the official app.
2. **Start the IMUs** — insert a CR2025 into each unit (pry the lid with a coin). A **red light blinks** when the IMU is on; it can take up to 30 seconds to appear.
3. **Sign in** to the app with your credentials.

4.3 · Pairing the IMUs

Open **Settings** (gear icon, top-right of the home screen) and go to **Sensors & Hardware**. Each body part has a **Pair** button; tap it and pick the sensor from the list. Identify a sensor by the last four digits of the serial number on its back. Once connected the row shows **Paired** in cyan with battery level and an **Unpair** option.



A Settings → Sensors & Hardware. Pair / Unpair each body part; "Head alignment" sits below.



B "Select a sensor to pair" — tap the matching serial number to connect it to that body part.

Colour-code standard — Right arm Blue · Trunk Green · Head Black · Left arm Red · Hand Silver · Forearm Beige. The optional HR chest band pairs with the mould-free sensor.

> **Head alignment.** The head IMU can face either way. In Settings, set **Logo side facing forehead** (logo forward) or **Back side facing forehead** (logo backward) to match how you fit it.

> **Test auto-pairing.** Log out, close and reopen the app — sensors should reconnect automatically. If you're not measuring right away, remove the batteries to save charge.

Phone settings (own phone)

- **Battery:** turn off adaptive/optimised modes; set the Wergonic App to **Unrestricted**.
- **Permissions:** Location and Nearby devices → **Allow all the time**.
- **Notifications:** enable for the Wergonic App.
- **Unused-apps / app-pausing:** disable for the Wergonic App.

4.4 · Create your tags on the dashboard

Sign in at **cloud.wergonic.com** and open **Tags** in the left menu. Tags organise your measurements in the cloud.

- **Activities** — choose *Work Task*, press *Add category*, then add activities (use *+ Add item* for several). Optional in online mode.
- **Work Cycles** — choose *Work Cycle*. Prepare them in Excel and import, or add them with the *+* button. Cycles are organised by Factory, Line, Workstation, Model, Task and Time. **Required** when measuring a work cycle.

> **Note.** Dashboard (web) screenshots are covered separately in the online help; this manual focuses on the mobile app.

5. Running a Measurement

5.1 · Prepare

- Fully charged phone and six paired IMUs
- Fresh CR2025 batteries (recommended)
- Clean Wergonic shirt & glove in the right size, plus a headband for the head IMU
- A ~2 kg weight for arm calibration

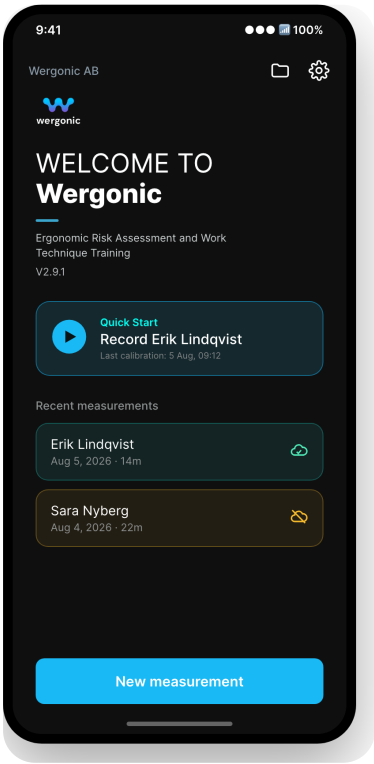
5.2 · Start the IMUs & fit the workwear

Insert batteries and confirm each IMU shows a **red blinking light**. Fit the sensors with the Movesense logo **facing outward and upright**:

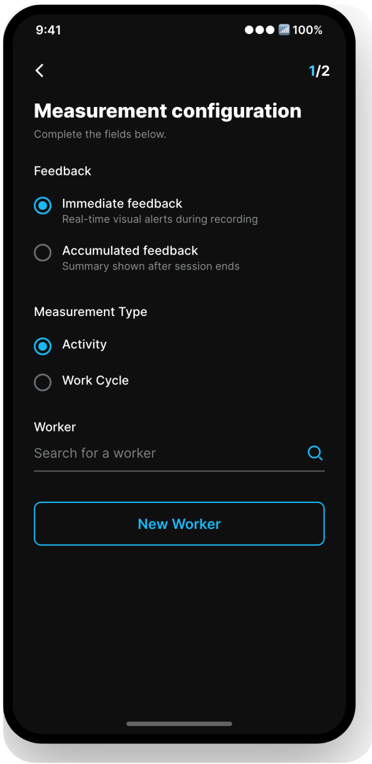
- **Shirt (trunk & arms):** slide IMUs into the pockets — Right arm Blue, Trunk Green, Left arm Red. Shirt should fit snugly.
- **Head:** attach the black IMU to the headband, logo forward or backward — match the Head alignment setting.
- **Glove (hand & forearm):** Hand Silver, Forearm Beige, in the matching pockets.

5.3 · Start a new measurement in the app

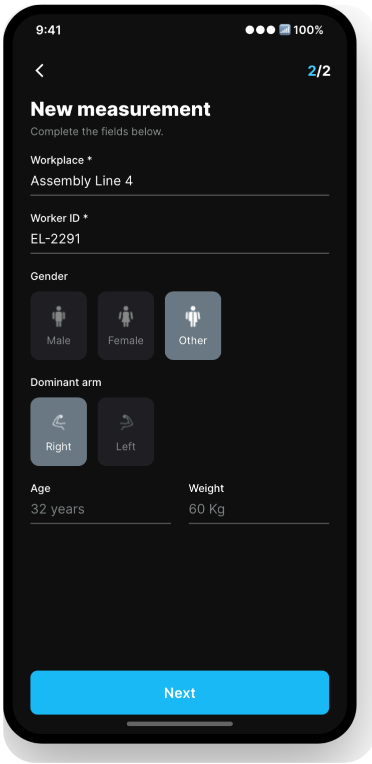
On the home screen tap **New measurement**, then complete the two setup pages and confirm pairing before calibration.



C Home — tap **New measurement**. The folder icon opens history; the gear opens settings.



D Page 1 — choose feedback type and **Measurement Type**: **Activity** or **Work Cycle**. Pick or search a worker.

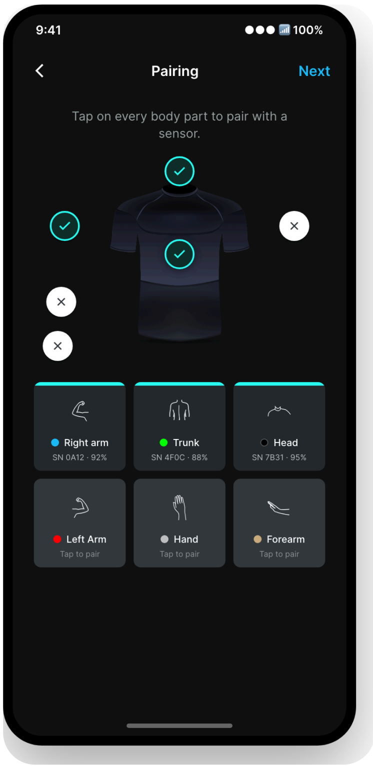


E Page 2 — worker profile (workplace, ID, gender, dominant arm, age, weight). Tap **Next**.

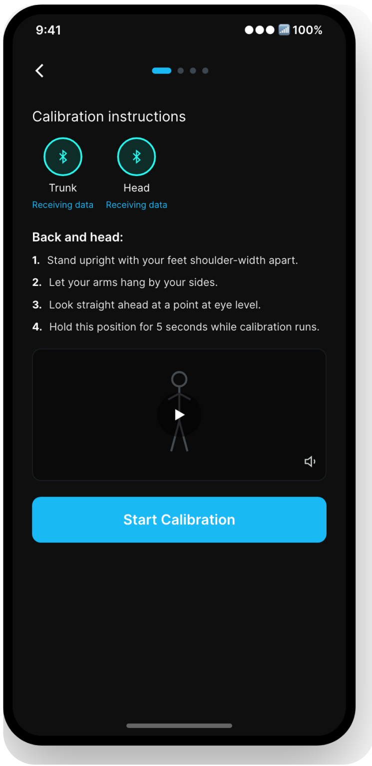
> **Online vs offline.** To load a pre-programmed **Work Cycle** or reuse a saved worker profile you must be online. Offline, create a new profile and sync later.

5.4 · Confirm pairing

The **Pairing** screen shows the shirt with each body part. Paired sensors show a **cyan check**; tap a body part to pair, or tap a paired one to unpair. When ready, tap **Next** (top-right) to calibrate.



F Pairing overview — connection cards show each sensor's serial and battery. Tap **Next** when all needed sensors are green.



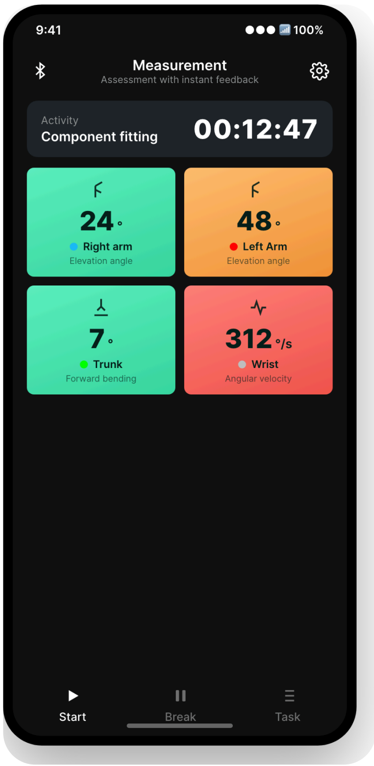
G Calibration — follow the numbered steps and demo video, then **Start Calibration** and hold the pose for 5 seconds.

5.5 · Calibrate the IMUs

Calibration is done per test person, in **four steps — Back and head**, **Right arm**, **Left arm**, and a **Forward reference**. Each step gives text, a demo video (with audio) and voice cues. The arm steps use the ~2 kg weight. Calibration is saved on the phone and can be reused for consecutive measurements on the same person **if the IMUs have not moved** — otherwise recalibrate.

5.6 · Live mode & recording

After calibration you enter live mode. Metric squares update in real time and colour by risk — **green** (safe), **orange**, **red**.



H Live mode — **Start** to record, **Break** to pause, **Finish** to end. Long-press a square to choose which metrics show; the gear (at 00:00:00) recalibrates a body part.

- **Choose metrics:** long-press any square, tick the metrics you want, then **Save** (top-left).
- **Recalibrate:** before starting (timer at 00:00:00), tap the gear → a body part.
- **Record:** tap **Start**; use **Break** to pause; tap **Finish** to end and save. When online you can also add activities during recording.

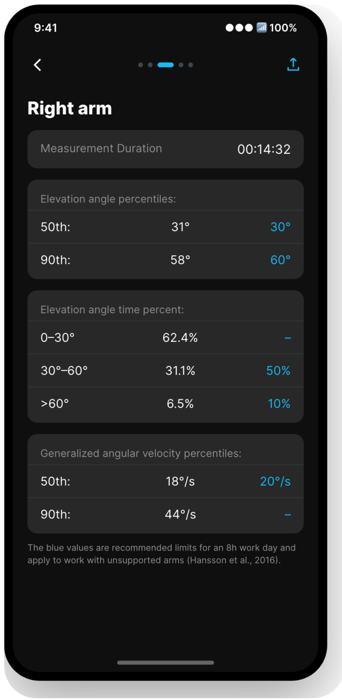
6. Reviewing Results

Review a session directly in the app, or in full on the web dashboard.

In the app

Tap the **folder icon** (home, top-right) to open your sessions. On a session, tap the ⋮ **menu → Report**. The report opens as **swipeable pages** — one per body part — with the numbers in tables. Reference limits are shown in blue.

- ⋮ menu also offers **Sync measurement**, **Share** and **Delete**.
- Tapping a session opens its **timeline** (activities, breaks).



In-app report — per-body-part tables; blue values are recommended 8-hour limits.

On the web dashboard

Online sessions upload automatically; offline sessions upload when you reconnect (may take a few minutes). Sign in at **cloud.wergonic.com**, open **Finished Sessions**, and click **Details** on a session. Results are shown as:

- **Pie charts** — posture (°) first, switch to *Speed of movement* for angular velocity (°/s). Export to Excel/PDF, or open *Show Detailed View* for time plots.
- **Tables** — posture, an algorithm-based reach estimate, then speed of movement.

7. Maintenance & Return

Between sessions

- **Remove the IMUs**, then machine-wash garments at **30 °C** with regular, perfume-free detergent (no disinfecting softener).
- **Disinfect** the removed IMUs.
- Avoid fully discharging the phone.

Returning the kit

- Clean the workwear as above.
- Batteries needn't be returned — recycle any you won't reuse.
- Fully charge the phone.
- Log out of Google Play and the Wergonic App.

8. Troubleshooting & FAQ

Troubleshooting

Problem	What to do
IMU won't blink / seems off	Moist fingers can discharge coin cells. Try a fresh battery; if it still won't start, contact Wergonic for a replacement.
IMU is slow to pair	Start the IMU *before* opening the app and wait ~1 minute. Don't repeatedly tap pair/unpair. Try another location — some environments disturb Bluetooth.
Battery lid won't open	Use only CR2025, inserted the right way up. Never use CR2032 — it can jam and damage the lid.
App stuck loading after calibration	Usually an offline network search. Enable flight mode *with Bluetooth on*, or work online for best stability.

Frequently asked questions

Do I need to be online to measure?

No — only Bluetooth is required. Data is stored on the phone and uploads when you're back online. Online mode is more stable and is recommended.

How far does Bluetooth reach?

Roughly 10 metres, but range depends on the environment where you measure.

> **Need help?** Contact the Wergonic team for hardware replacements or support with setup and measurements.